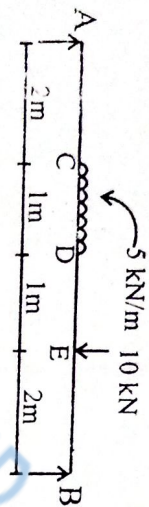


Q6 For the loading shown in fig.1 find the position and magnitude of max. deflection. Also find deflection under point C, D & E. Flexural rigidity = $9 \times 10^{13} \text{ N-mm}^2$



20

Q7 Write short notes:

- Area moment method
- Rankine formula for columns
- Shearing stress in beams
- Strain energy

(4x5=20)

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.

B.E. II (M)

4

M.S.

SECOND B.E. (MECHANICAL ENGINEERING)

III SEMESTER EXAMINATION -2019

ME-204 A: MECHANICS OF SOLIDS (M)

Time - Three Hours

Maximum Marks - 100

Note:- (1) Attempt any FIVE questions.

- All questions carry equal marks.
- Marks allotted to each part of the question are indicated on the right side.
- Draw neat sketches, wherever necessary.
- Assume any data, if not given and mention it.

1. (a) Derive the expression for elongation of a bar of uniform section due to self weight. 6

(b) A 15 mm diameter steel rod passes centrally through a copper tube of 60 mm external and 50 mm internal diameter. The tube is closed at each

end by rigid plates of negligible thickness. The nuts are tightened lightly on the projecting parts of the rod. If the temperature of the assembly is raised by 60°C , calculate the stresses developed in Copper and steel.

$$E_s = 2 \times 10^5 \text{ N/mm}^2, E_c = 1.05 \times 10^5 \text{ N/mm}^2$$

$$\alpha_s = 12 \times 10^{-6} / ^{\circ}\text{C}, \alpha_c = 17.5 \times 10^{-6} / ^{\circ}\text{C}$$

14

- 2 (a) Prove the following relation among elastic constants:

$$(i) E = 2C(1 + 1/m) \quad (ii) E = 3K(1 - 2/m)$$

8

- (b) A copper cylinder having 45 mm internal diameter and 1.5 mm wall thickness is closed at its ends by plugs which are at 450 mm apart. The tube is subjected to internal pressure of 3 MPa. Calculate,

(i) Longitudinal stress (ii) Change in length

(iii) Change in diameter (iv) Change in volume

12

- 3 (a) Explain the theory of pure bending.

6

- (b) A 4m long beam with rectangular section of 100 mm width and 200 mm depth is simply supported at the ends. If it is load with UDL of 4 kN/m

3695

2

(Contd.)

throughout the span and a concentrated load $P = 2 \text{ kN}$ placed at a distance of 1.5 m from one end, determine max. bending stress.

14

$$4 \quad (a) \quad \text{Derive } \frac{T}{J} \cdot \tau = \frac{C\theta}{L} = \frac{q}{r}$$

8

Also mention the assumptions.

- (b) A Steel bar of rectangular section $30 \times 40 \text{ mm}$ with one end fixed and other is kept free. The bar is 1.75m long. Determine the buckling load and the corresponding stress using Euler's formula. Also determine the minimum length for which Euler's equation may be used to determine the buckling load, Take $E = 2 \times 10^5 \text{ N/mm}^2$ and proportional limit of the material is 200 N/mm^2 .

12

5

- (a) Define principal stresses and principal planes.

4

- (b) A rectangular block of material is subjected to a tensile stress of 200 N/mm^2 on one plane and a tensile stress of 100 N/mm^2 on a plane at right angles, together with shear stress of 120 N/mm^2 on the same plane. Find

- (i) The direction of principal plane
(ii) Magnitude of principal stresses
(iii) The magnitude of max. shear stress and
(iv) Plane of max. shear stress

16

3695

3

(Contd.)

per unit volume to be $\tau^2/3G$ where G is modulus of rigidity. Also find the inner and outer diameters if the shaft transmits 5MW at 120 rpm when energy stored is 35 kN/m^3 .

Assume modulus of rigidity $G = 78 \text{ GPa}$.

10

6. (a) Differentiate between columns and struts. Also state Euler's theory and its assumptions for analysis of struts and columns. 10

(b) A 3 m long cast iron hollow column with both ends firmly fixed supports on axial load of 400 kN the inside diameter of the column is 0.6 time the outside diameter. Determine the section of the column. Assume factor of safety to be 5, crushing stress as 560 MPa and Rankine's constant as $1/1600$. 10

7. Write short note on :-

- (i) Principal stresses & principal strain
- (ii) Mohr's strain circle
- (iii) Various methods to determine slope and deflection of beams.
- (iv) Max normal stress theory of failure. 4x5=20

8230

4

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 15ME038515

B.E. II (M)

4

M.S.

SECOND B.E.

(MECHANICAL ENGINEERING)

III SEMESTER EXAMINATION - 2017

ME-204 A : MECHANICS OF SOLIDS (M)

Time - Three Hours

Maximum Marks : 100

Note :- (1) Attempt any Five questions.

- (2) All questions carry equal marks.
 - (3) Marks allotted to each part of the question are indicated on the right side.
 - (4) Draw neat sketches, wherever necessary.
 - (5) Assume any data, if not given and mention it clearly.
1. (a) Define Poisson's ratio and evaluate linear and volumetric strains. 5
 - (b) Establish relation amongst various elastic constants. 5
 - (c) A steel bar 2400 mm long is 18mm in diameter for half the length and 12mm diameter for the other half length. Find- 5

8230

1

(Contd.)

(i) The extension of the bar when subjected to a tensile force of 22.5 kN

(ii) For a steel bar of uniform diameter having same length and volume, what would be extension when subjected to same tensile force?

Assume modulus of elasticity as 200 GPa. 10

2. (a) A steel rod 18 mm in diameter passes through a steel tube 25 mm in internal diameter and 30 mm in external diameter. The tube is 750 mm long and is closed by rigid washers of negligible thickness which are fastened by nuts threaded on the rod. The nuts are tightened until the compressive load on the tube is 20 kN. Calculate stresses in the tube and the rod.

Also find increase in these stresses when one nut is tightened by one quarter of a turn relative to the other. There are 4 threads per 10 mm. Assume $E = 200 \text{ kN/mm}^2$.

(b) Compare the maximum tensile stresses of a thin cylinder to that of a thin spherical shell having same internal pressure and d/t ratio where d and t are the internal diameter and thickness respectively. Also find ratio of their proportional increase in volumes. Assume Poisson's ratio as 0.3. 10

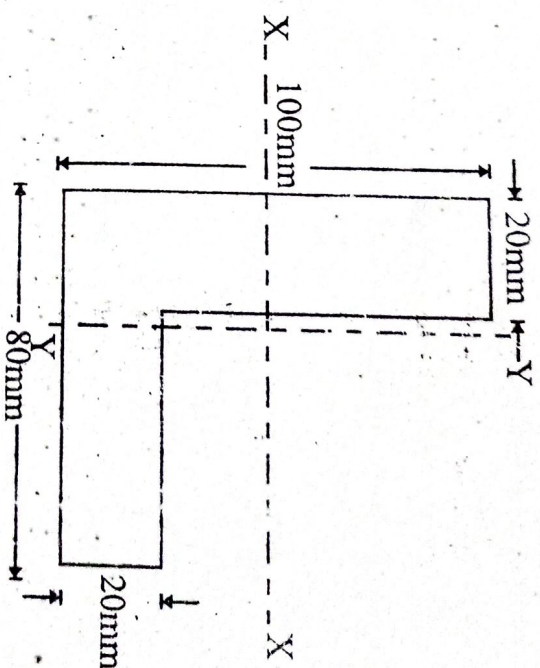
3. (a) A timber beam is freely supported on supports 6 metres apart. It carries a uniformly distributed load of 12 kN per metre run and a concentrated load of 9 kN at 2.5 metres from the left support. If the maximum permissible stress for the beam is limited to 8 MPa. Design suitable section for the beam.

Assume depth of the beam as twice of its width.

(b) A simply supported beam has a distributed load of linearly varying intensity with zero at one end to w per unit run at the other end. Draw shear force and bending (Contd.)

diagrams for the beam.

4. (a) Find the moment of inertia about the centroidal axis XX and YY for the section shown in fig 1 10



(b) A beam 100 mm wide and 150 mm deep supports a uniformly distributed load over a span of 2 metres. If the safe stresses are 28 MPa longitudinally and 2 MPa in transverse shear. Calculate maximum load which can be supported by the beam. 10

5. (a) A 80 mm wide and 180 mm deep cantilever is having a span of 3m. It carries a uniformly distributed load of 6 kN/m intensity on a 2m length of span starting from the free end. Determine the slope and the deflection at the free end.

Assume modulus of elasticity as 205 GPa. 10

(b) Determine the ratio of inner to outer diameters of a hollow shaft that attains a maximum shear stress of τ when subjected to pure torque. Take strain energy

(Contd.)

A beam is of square section of side 'b'. find the ratio of the flexural strengths of the section when the beam is placed such that (i) two sides are horizontal (ii) One diagonal is vertical. 6

- Q5 (a) A beam of length l has one support at the left end and the other support being at a distance $l/3$ from the right end. The beam carries a uniformly distributed load of w per unit run over the whole length. find the deflection at the right end of the beam. 6

- (b) A shaft has to transmit 105 kw power at 160 rpm. If the shear stress is not to exceed 65 N/mm² and the twist in a length of 3.50 metre must not exceed 1°. Find a suitable diameter. Take $C = 8 \times 10^4$ N/mm² 6
- Q6 (a) A hollow cast iron column whose outside diameter is 200 mm has a thickness of 20 mm. It is 4.5 metre long and is fixed at both ends. calculate the safe load by Rankine's Formula using a factor of safety of 4. Compare the Rankin's load with Euler's load. Take

$$f_c = 550 \text{ N/mm}^2, \alpha = \frac{1}{1600} \text{ and } E = 8 \times 10^4 \text{ N/mm}^2. \quad 6$$

- (b) A Cylindrical shell 900 mm long 150 mm internal diameter, having a thickness of metal 8 mm is filled with a fluid at atmospheric pressure. If an additional 20000 mm³ of fluid is pumped into the cylinder find (i) the pressure exerted by the fluid on the cylinder (ii) the hoop stress induced Take $E = 2 \times 10^5$ N/mm² and $1/m = 0.3$ (6)

Q7 Write short notes on any two-

6 × 2 = 12

- Theories of failure
- Beam of Uniform strength
- Stress strain diagram

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. J.S.M.E.C.

2.5.5

B.E. II (M)

4

M.S.

Second B.E. Examination 2016 (Mechanical Engineering)

(III- Semester Examination Scheme)

ME-204 A : MECHANICS OF SOLID (M)

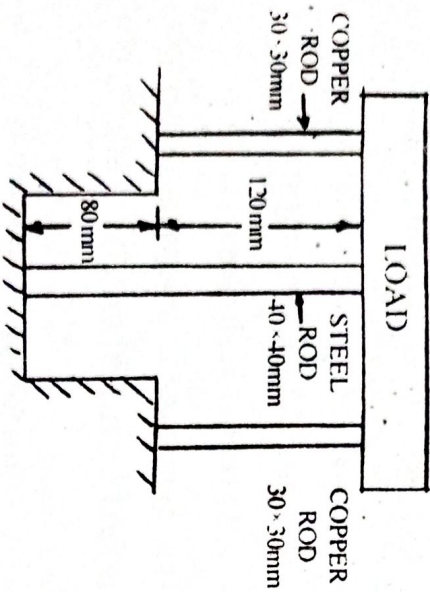
Time Allowed - Three Hours
Maximum Marks - 60

Note : (1) Attempt any Five Questions.

- All questions carry equal marks.
- Marks allotted to each part of the question are indicated on the right side.

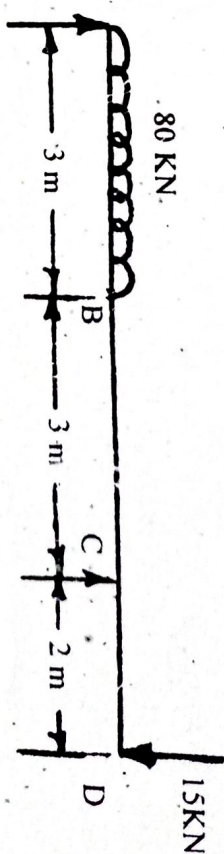
- Draw neat sketches. Wherever necessary.
- Assume any data, if not given and mention it clearly.

- 1 (a) A steel tie rod 40 mm in diameter and 2m. long is subjected to a pull of 80 kN. To what length the bar should be bored centrally so that the total extension may increase by 20% Under the same pull, the bore being 20mm. diameter. Take $E = 2 \times 10^5 \text{ N/mm}^2$ 6
- b) Two copper rods and one steel rod together support a load as shown in fig 1. if the stresses in copper and steel are not to exceed 60 N/mm^2 and 120 N/mm^2 respectively. Find the safe load that can be supported. Young's Modulus for steel is twice that of copper. 6

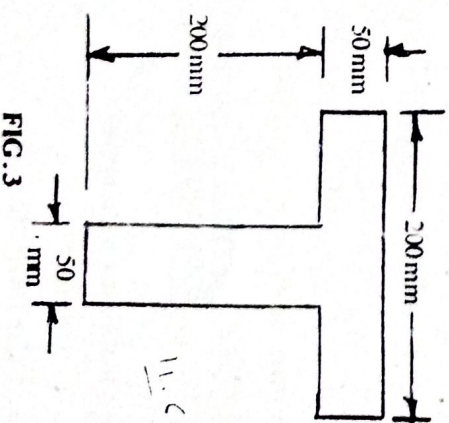


- 2 (a) A bar of steel is of square section 60 mm x 60mm and 180 mm long. It is subjected to an axial compressive load of 300 kN. The lateral strain is prevented by the application of uniform external pressure. if $\nu = 0.3$ and $E = 2 \times 10^5 \text{ N/mm}^2$. Find the alteration in the length of the bar. 6
- b) Two Planes which are at right angles carry shear stresses of intensity 17.5 N/mm^2 While these planes also carry

- a tensile stress of 70 N/mm^2 and a compressive stress of 35 N/mm^2 respectively. Determine the principal planes and the principal stresses. Also determine the maximum shear stress and the planes on which it acts. 6
- Q3 Construct the bending moment and shear force diagrams for the beam shown in fig-2. Determine the point of contraflexure if any. 12



- Q4 (a) The T-shaped cross section of a beam shown in fig.3 is subjected to a vertical shearforce of 100 kN. Draw the shear stress distribution diagram for the section and find the maximum shearing stress across the section. 6



"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 13.ME.S. 4076

B.E. II (M)

4

M.S.

Second B.E. Examination, 2015

(Mechanical Engineering)

(III-Semester Examination Scheme)

ME-204 A : MECHANICS OF SOLID (M)

Time—Three Hours

Maximum Marks—60

Note :— (1) Attempt any FIVE questions.

(2) All questions carry equal marks.

(3) Marks allotted to each part of the question are indicated on the right side.

(4) Draw neat sketches, wherever necessary.

(5) Assume any data, if not given and mention it clearly.

1. (a) A copper rod ABCD of 800 mm^2 cross-sectional area and 7.5 m long is subjected to forces as shown

in Fig. 1. Find the total elongation of the bar. Take modulus of elasticity as 100 GPa. 6

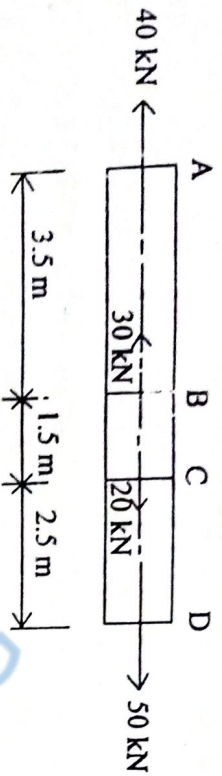


Fig. 1

(b) A steel block of 50 mm side is subjected to a force of 6 kN (Tension), 8 kN (Compression) and 4 kN (Tension) along X, Y and Z direction respectively. Determine the change in volume of the block. Take E as 200 GPa and Poisson's ratio μ as 0.3. 6

2. (a) The intensity of resultant stress on a plane AB (Fig. 2) at a point in a metal under stress is 80 N/mm^2 and is inclined at 30° to the plane AB. The normal component of stress on another plane AC is 60 N/mm^2 . Calculate the following :
- Principal stresses and their directions
 - Maximum stresses and their plane.

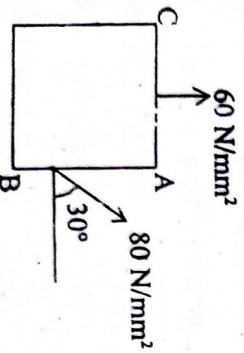


Fig. 2

(b) A cylindrical shell is 3 m long, 1 m internal dia and 15 mm thick. Calculate the maximum intensity of shear stress induced and also the changes in the dimensions of the shell if it is subjected to internal pressure of 1.5 N/mm^2 . Take $E = 200 \text{ GPa}$ and $\mu = 0.3$. 6

3. A beam ABCDE is supported at A and B and has overhang BE. Draw the shear force, bending moment and axial force diagram. Also determine the point of contraflexure. 12

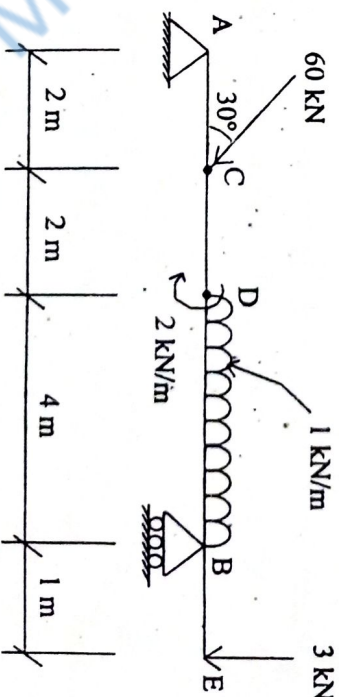


Fig. 3

4. A beam ABCD, as shown in Fig. 4 is loaded. The beam has an I-section, throughout its length, of $350 \text{ mm} \times 200 \text{ mm}$. The web and flange thickness are 12.5 and 25 mm respectively. Determine the bending stress and

sketch its distribution. Also determine the distribution of shear stress when it is critically loaded. 12

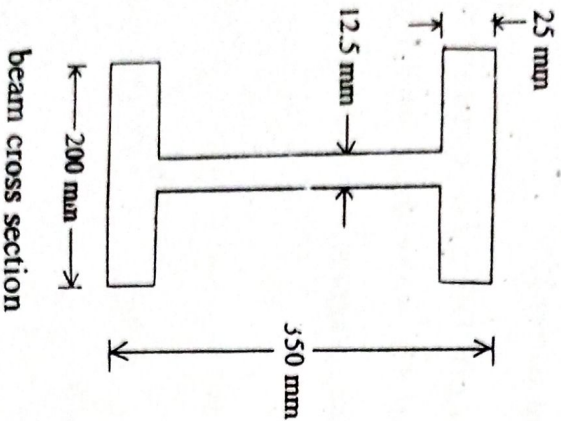
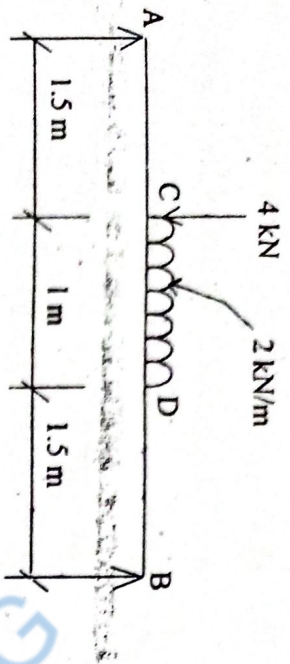


Fig. 4

5. (a) Derive the formula for slope and deflection at the free end of a cantilever beam subjected to point load W at the free end. 6
- (b) Derive Euler formula for buckling of a column supported by hinge at both ends. The column is subjected to loaded axially. 6

6. (a) For a circular shaft subjected to torque T , derive the following relationship :

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{\ell}$$

where J = polar moment of inertia

τ = shear stress

r = radius

G = modulus of rigidity

θ = angular twist

ℓ = length of shaft

- (b) A shaft is transmitting 100 kW at 180 rpm. If the allowable shear stress in the material is 60 MPa, determine the suitable diameter of shaft. Also determine whether the shaft is able to resist an angular twist of 1° in 3 m length, take $G = 80$ GPa. 6

7. Write short notes on any **THREE** of the following :

- (i) Relation between elastic constants
- (ii) Mecauly's method
- (iii) Empirical formula for column
- (iv) Mohr's circle for stresses. 4×3=12

- (b) A bar of steel is of square section 60mm x 60mm x 180mm long. It is subjected to an axial compressive load of 300KN. The lateral strain is prevented by the application of uniform external pressure. If $\mu = 0.3$ and $E = 200 \text{ KN/mm}^2$. Find the change in the length of bar. (6)

6. A transmission shaft supported by two bearings B₁ and B₂ carry two pulleys A and C as shown in Fig. 4 20 KW pulley at 500rpm is transmitted from pulley A to C. The shaft is made of steel having allowable stress 100 N/mm^2 . Determine the diameter of the shaft on strength basis if the diameter of pulleys A and C are 200mm and 350mm respectively and the ratio of tension is 2.0. (12)

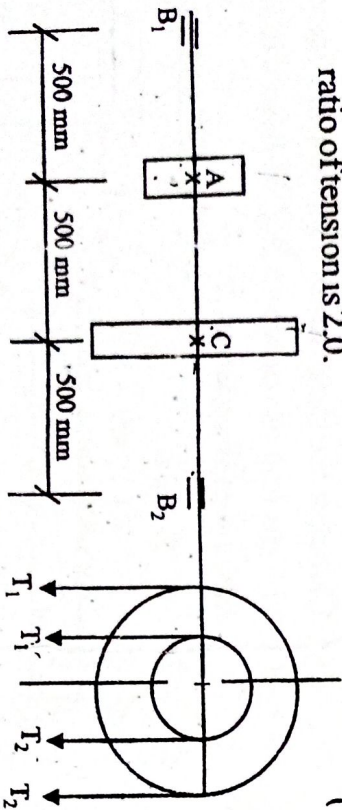


Fig 4

Write short notes on ANY FOUR of the following:

- Stress strain curve for mild steel
- Empirical formula for column
- Mohr's circle for strain
- Theories of failure
- Area moment method

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 4202307...

B.E. II (M)

M/c Drawing

Second B.E. Examination, 2014

(Mechanical Engineering)
(III - Semester Examination Scheme)

ME - 204 A : MECHANICS OF SOLID (M)

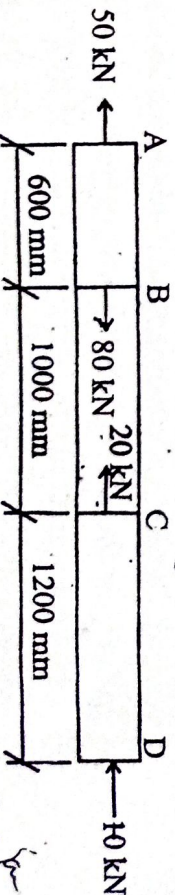
Time : Three Hours

Maximum Marks : 60

Note : (1) Attempt any FIVE questions

- All questions carry equal marks.
- Marks allotted to each part of the questions are indicated on the right hand side.
- Draw neat sketches, wherever necessary.
- Assume any data, if not given, and mention it clearly.

A brass bar having a cross-sectional area of 1000 mm^2 is subjected to axial forces as shown in Fig. 1. Find the total change in length of the bar. Take $E = 1.05 \times 10^5 \text{ N/mm}^2$. (6)



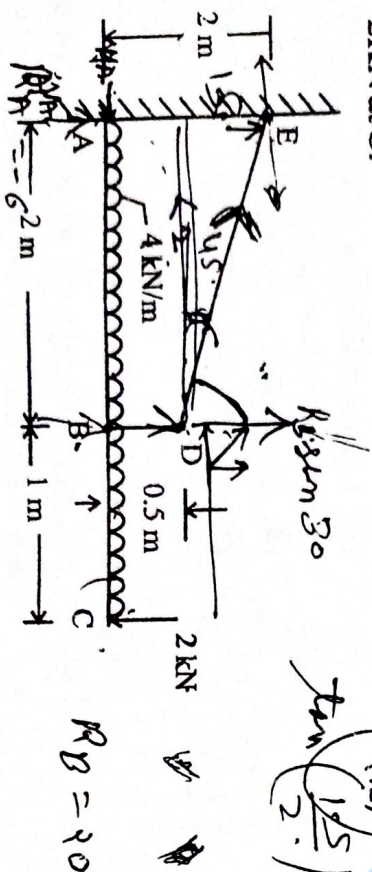
(b)

A point in a elastic material under stress is subjected to normal stress 50 N/mm^2 (tensile) along X axis and 30 N/mm^2 (compressive) along Y axis. The element is subjected to shear stress of magnitude 25 N/mm^2 . Find the principal stresses and position of principal planes. Also find the maximum shear stress and position of its plane. Use Mohr's circle method.

(6)

A beam ABC shown in Fig 2 hinged to the wall at A.

A vertical bracket BD is firmly fixed to the beam at B and a tie DE is hinged to the bracket at D and to the wall at E. Draw shear force and bending moment diagram for the beam ABC when it carries a Udl of 4 KN/m run over the whole length and a point load of 2 KN at C.



(3) (a)

A steel beam is simply supported at the ends on a span of 8 m and carries Udl of 8 KN/m on the whole span. In addition, a connection made to the beam at 5 m from the left end exerts a downward point load of 80 kN together with a clockwise couple of 60 kN m acting in the plane of bending of the beam.

Determine the location and magnitude of maximum deflection, I_{xx} for the beam section $= 4.8 \times 10^8 \text{ mm}^4$ and $E = 200 \text{ KN/mm}^2$.

(6)

(b) Derive an expression for hoop stress developed in a thin cylinder when it is subjected to internal pressure $P \text{ N/mm}^2$

(4)

4. (a) Stating the assumptions made in the theory of simple bending, derive the following relation:

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

Where M, I, σ, y, E and R have their usual meaning.

(6)

(b) Calculate the shear stress distribution over the beam of T-section as shown in Fig. 3. Also calculate the maximum shear stress.

(6)

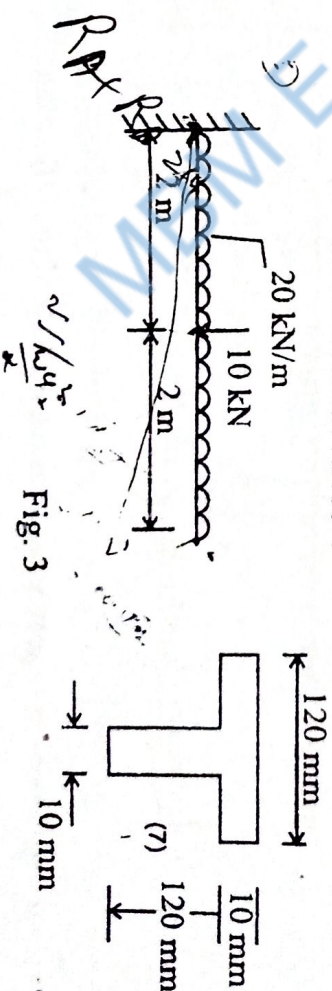


Fig. 3

(5) (a)

Derive Euler formula for buckling of a column fixed at one end and free at the other end. Column is subjected to loaded axially.

(6)